

Task 4:

What differences did you observe?

The baseline model was trained on gesture data collected from many different users, so it recognized generalized versions of each digit. For some digits it worked well, but for others it either misclassified or failed to recognize the gesture consistently likely because my drawing style differed from the training data.

The fine-tuned model was trained on my own 10 strokes per digit, so it learned my specific way of drawing each number. Recognition felt more consistent for digits that I drew in an unconventional way.

Which model performed better and why?

The fine-tuned model performed better overall for my personal gesture style. Because it was trained directly on my data, it adapted to my speed, stroke direction, and hand movement patterns. The baseline model had no exposure to my specific style, so it relied on generalizations that didn't always match.

Did fine-tuning help for your specific gesture style?

Yes, fine-tuning helped particularly for digits that are ambiguous in shape (like 1 vs 7, or 3 vs 8), where small stylistic differences matter. However, because the fine-tuning dataset was very small (only 10 samples per digit), the model can overfit to one specific way of drawing and fail if the gesture varies slightly.